

Roadmap: deciding whether the field earns its keep

2026-08-27, evening. Jonas asked for a plan covering the two decidable claims in the assessment ("the accumulator closes the within-scenario gap"; "the level transfers across scenarios") and, more broadly, for rolling out everything in the assessment's ADAS-trigger section. This document is that plan. It also records the answers Jonas gave to the fitting plan's four questions and what they change. Companion documents: docs/czb_fitting_plan.md (the model and estimation machinery), docs/active_inference_for_czb_assessment.md (the argument this plan tests).

0 What was already run tonight, because it was cheap

- **The overfitting question is answered.** The within-scenario comparison is not an in-sample artifact: under leave-one-criticality-out cross-validation (fit on two criticality levels, predict the third) the field gives out-of-sample corr 0.864 / RMSE 0.147 against the design regressions' 0.934–0.949 / 0.097–0.108, and leave-one-timepoint-out gives the same picture (field 0.870, regressions 0.933–0.948). The regressions genuinely generalize better *within this scenario*, including when extrapolating to an unseen criticality level. The field's case therefore rests, exactly as stated, on the accumulator, on portability, and on state-based deployability — not on surface fit.
- **The perceived-safety correlation exists and is strong** (Jonas's request; figure figures/ps_vs_field.png, script replication/czb/ps_vs_field.py). The stored PS scale is perceived *unsafety* (dictionary gotcha 1: higher = less safe). Against the field: cell-level Spearman $\rho = 0.93$ (Pearson $r = 0.84$) over the 18 cells, trial-level $\rho = 0.58$, and within-criticality trial-level $\rho = 0.64 / 0.52 / 0.39$ (TTC4/6/8). The ratings reproduce the same time-compression mismatch as the intervention surface: ratings keep climbing through the window where the static field flattens — a third independent appearance of the accumulator's signature (after P(intervene) and the ratings' shape, the button densities will be the fourth).

0b Amendments from the same evening's discussion

Jonas's follow-up questions changed four things; recorded here so the stages below read correctly.

- **The named comparator is now his 2D state rule, not the design regression.** A probit in 1/TTC and lateral offset — both observable state variables — reaches held-out corr 0.921 on the cut-in surface (field 0.864; design regression 0.934, demoted to an upper-bound reference since its covariates are experiment clocks, not state). Two things should be said about the 2D rule: within this stimulus family the lateral-offset trajectory is nearly identical across criticality levels, so it spans almost the same space as the design regression; and its second observable is scenario-specific (lane-crossing distance for cut-in, PET for LTAP, clearance for overtakes), which is exactly the conventional per-scenario route whose cross-scenario version is the elliptical formulation. The transfer test is therefore a three-way comparison: one-scalar field, per-scenario 2D rules, elliptical joint.
- **The comfort and dread levels are both fitted, not fixed at 4 and 8.** Jonas's correction on the DZB's meaning is adopted: the dread boundary is the deceleration drivers do not voluntarily push beyond, a behavioral limit rather than the physical $a_{\max}$ — so 8 m/s^2 is an upper anchor, not the answer, and 4 may also be high for comfort. The ordered model fits both levels as free parameters per the hierarchy; the 4/8 pair survives only as the staging default for computing example fields.
- **A secondary model fits the conventions.** $a_{\text{OV,min}}$ and t_{react} (the driver's *assumed* worst case and budget — distinct from motor latency) can in principle be identified at population level, because they enter the boundary with different kinematic signatures (t_{react} scales with v_{ego} , $a_{\text{OV,min}}$ with the lead's stopping distance, a_{allowed} with v_{react}^2) and the scenarios span 5.5–30.5 m/s. The primary analysis keeps them fixed for falsifiability; the secondary lets the data place them, and disagreement between the two is itself informative.
- **Epistemic value gets reconsidered on evidence, not dropped for convenience** (stage E below). The deposit contains the authors' own $\alpha = 0$ runs: rear-end 28 matched configurations, collision difference exactly 0.000; intersection 6 matched, within 0.005; **oncoming 3 matched, collisions 5.7 points lower with the term on.** Inert in the longitudinal scenarios, possibly useful in the lateral one (replication/czb/epistemic_ablation_check.py).

1 Stage A — close the within-scenario question (1–2 sessions, all data in hand)

- 1 **Synthetic-recovery harness** for the stage-1 hierarchical model (fitting plan §3). Simulate from known parameters, refit, confirm recovery. Gate for everything below.

- 2 **Stage-1 hierarchical fit** on the Random cut-in surface: per-driver c_i , the ordered braking-expectation thresholds, and **both bias variants** — group-level b and hierarchical b_i with shrinkage — compared by held-out likelihood and by posterior predictive on the C1 cells (the implications question; section 5.1 below).
- 3 **The accumulator layer**, fitted on the same surface. Decision criterion, stated in advance: the accumulator variant should recover most of the gap to the design regressions' cross-validated RMSE (~ 0.10) without any per-scenario parameter. If it does not, that is recorded as evidence against the framework, per the assessment.
- 4 **Button-side validation**: predict the press-time densities (2 396 trials) with one paradigm shift δ , fitted both as a level shift and as a rate shift; the shape decides. Motor latency fixed per section 5.3.

2 Stage B — the transfer test on data we already hold (the gate)

The Random design contains, with the same 43 participants and PS ratings throughout: cut-in ($3 \times C1-C6$), LTAP (8 PET levels $\times$ 2 speeds), cyclist overtake (3 clearances $\times$ C1-C5), truck overtake (5 lateral offsets). This is exactly the transfer design, held in-house. What blocks it is field construction, not data:

- 1 **Cyclist overtake** (first, easiest): same-lane longitudinal geometry, roles fixed by instructed vehicle rather than lateral span (the loader mis-assigns these — review finding 1.5). The ego closes on a cyclist; the existing rear-end field applies nearly unchanged.
- 2 **Truck overtake / truck lateral**: the ego passes a stationary or drifting truck with lateral clearance as the criticality axis. Needs the lateral-clearance term rather than the car-following counterfactual; the preference function's lateral machinery (p_{lat} , and P_{lane} run "sideways") covers it, but this is a construction to argue, not assume.
- 3 **LTAP**: crossing geometry — the cut-in x/y conventions are meaningless; the collision-relevant quantity is arrival-time separation at the conflict zone. This is the intersection scenario's world model; the construction should follow the authors' intersection scenario the way the cut-in followed rear-end.
- 4 **The test itself**: fit c (and the accumulator) on cut-in only; predict the other three response surfaces with everything frozen. Pre-registered readout: predictive correlation per scenario against (a) chance, (b) a per-scenario refit ceiling, and (c) the elliptical fallback fitted on conventional indicators. Pre-registered confound control: the per-scenario *instruction* differs (intervene / abort / yield), so a uniform shift per scenario is permitted in a secondary analysis and the primary analysis allows none — both reported.
- 5 **A cheap preliminary before any of that**: the PS ratings exist for all four scenarios. Once a scenario's field exists, the cell-level rating-versus-field correlation (as run tonight for cut-in) is a transfer check that needs no fitting at all. I would run it per scenario as each field construction lands.

Test-track data (Jonas's offer). What would make external LTAP/OD data usable, in order of importance — this matches `docs/data_requirements.pdf`, which was written for exactly this handoff: (1) paired-vehicle kinematics at ≥ 10 Hz (positions and speeds of both vehicles on a common clock and datum; headings if turning); (2) a response-onset label per trial (brake onset, or release/steer onset, extracted however the study defined it — the definition matters more than the choice); (3) a per-driver identifier across trials, since the whole point is per-driver levels; (4) the staging protocol (scripted speeds, PET/gap levels, instruction wording — the instruction confound again); (5) vehicle dimensions. Nice to have: gaze or distraction state, and any subjective ratings. With (1)–(3) the test-track data can serve as a *second population* for the percentile question, not merely a replication.

3 Stage C — the percentile sensitivity analysis (cheap; run right after stage 1)

From the fitted population distribution of c : compute the implied trigger onset time per stimulus for percentiles 50–95 in steps of 5, and report $d(\text{onset})/d(\text{percentile})$ in seconds alongside the estimation uncertainty of the percentile itself. Decision this informs, stated in advance: if the trigger moves more per 5 percentile points than per sampling-uncertainty band, the bottleneck is the percentile *choice* (a policy question), not the estimator, and further estimator refinement is deprioritized. The same sweep run on the comfort-versus-dread axis ($a_{allowed}$ from 3 to 8 m/s^2) quantifies which of the two choices dominates.

4 Stage D — the comfort-versus-safety audit (the causation synergy)

For a small grid of candidate triggers (percentile $\times$ $a_{allowed}$), replay the QUADRIS ensemble with an intervention fired at the trigger's level crossing and score counterfactual outcomes with the existing causation machinery (crash probability, injury-weighted severity, and — for the nuisance side — trigger rate in the non-crashing seed majority). Output: a two-axis map (safety benefit versus nuisance frequency) with the CZB percentiles marked on it. This is the concrete version of "trigger set by comfort, audited by crash outcome", and it directly tests the QUADRARUM

document's ~0.1 s observation on our own ensemble. Effort: moderate — the runner, weights and metrics all exist; the new piece is the intervention injection.

4b Stage E — planning and epistemic value, reconsidered on evidence

The field method's omission of expected-free-energy planning is not purely convenience, and the distinction matters. What the field *does* include is the predictive rollout of the **other** vehicle — the lane-entry weight is exactly the closed-form expectation of the conflict geometry under the target's continued motion. What it omits is rollout of the **ego's own policies**. For the study-1 stimuli that omission is arguably correct rather than approximate: the instruction asks when doing nothing stops being acceptable, which is the pointwise deficit of the no-action trajectory — the thing we compute. Where policy rollout plausibly matters is where scenarios differ in escape affordances, which is precisely the transfer set (LTAP: yield or proceed; overtake: abort or continue). Three bounded experiments, each with a decision rule:

- 1 **E.1 — closed-form versus sampled expectation** (small): Monte Carlo the released binary gates under the model's own steering noise along the cut-in clips and compare with `P_lane`. If the closed form tracks the sampled expectation within the data's resolution, the "field = rollout in closed form" reading is validated quantitatively rather than argued.
- 2 **E.2 — an uncertainty term where the data can see it** (moderate): the truck and car cut-ins are matched on TTC levels but differ hugely in occlusion and apparent size. If truck responses at matched kinematics are systematically earlier than any kinematic covariate explains, an epistemic/precision term earns a place in the field; the Button truck trials are the test bed. This is also the unexercised gaze half's natural entry point.
- 3 **E.3 — a policy-set field for the transfer scenarios** (moderate): for fixed clips, EFE over a small discrete policy set (a few braking levels; the instructed alternatives) is closed-form kinematics, not CEM — cheap. Comfort becomes "the best available policy is still acceptable". Decision rule: if the per-scenario instruction shifts (stage B.4's secondary analysis) shrink materially when the policy set replaces the free shift, planning has earned its way back in; if not, the pointwise field stands.

The ablation result in section 0b sits behind all three: epistemic value is inert in the authors' longitudinal runs, hints at usefulness in the lateral one, and our $\alpha = 0$ matches an ablation they ran themselves — so nothing above contradicts the validated base; it extends it where the evidence points.

4c What the NDS access is for

Four distinct uses, in the order I would take them:

- 1 **Estimating the paradigm offset δ empirically.** Fit `c` on naturalistic brake onsets in lead-vehicle and cut-in events (the calibration machinery exists and needs only kinematics — `comfortzone.calibrate`), and compare with the study-1 estimates. The difference *is* the paradigm-plus-context offset, measured rather than assumed — the single biggest upgrade available for the absolute-trigger question.
- 2 **Exposure denominators for stage D.** How often routine driving crosses a candidate trigger level determines the nuisance rate — the deposit ensembles cannot provide this; NDS is the only source that can.
- 3 **A percentile consistency check.** The distribution of self-selected margins in routine driving (steady-state THW at speed, accepted lateral clearances) against the fitted comfort-level distribution: drivers should mostly live inside their own comfort zones, and the fraction who do not is a direct calibration check.
- 4 **Transfer events in the wild** — naturally occurring cut-ins and LTAP conflicts as a held-out scenario set with no instruction at all.

5 The four questions: answers received 2026-08-27, and what follows

5.1 Bias: test both, and why I leaned toward shrinkage. Agreed — both variants run in stage A.2 and the data decides. The reasoning behind my lean, as requested. A group-level bias asserts every participant shares one false-alarm propensity; a per-driver unshrunk bias estimates each from ~12 pre-onset trials, where a driver with 1 press in 12 gets $b = 0.083$ largely made of noise. Hierarchical shrinkage is the middle position: each driver's estimate is pulled toward the group mean in proportion to how little data supports it, so with thin data it *behaves* like the group model except for genuinely extreme drivers. The reason I think the driver dimension is worth keeping at all is a specific leakage risk: response bias and boundary level are plausibly correlated across people (a trigger-happy participant both false-alarms at C1 and crosses early at C5). Under a group-level bias, that person's excess C1

responses have nowhere to go except *into their fitted c_i* , biasing it low — and since the deliverable is a population *percentile*, distortion concentrated in the tail people is exactly the distortion that matters. The empirical implications to check when both are run: the group model should show systematic C1 misfit for the extreme participants, and, if the leakage is real, a wider fitted σ_c than the shrinkage model. If neither appears, group-level wins on parsimony and we say so.

5.2 The braking-expectation question. Per Jonas: treat C2B_2 as the human decision (own braking) for now; a new dataset framed that way is coming and will supersede this reading. Consequence for stage A: the ordered model still fits two nested levels; only the *interpretation* of the upper level (dread as "situation demands hard braking") is held provisionally rather than asserted, and the write-up should flag it until the new data lands.

5.3 Motor latency: fixed at 0.25 s, with the rationale. The accumulator models the decision, so λ should cover only post-decision motor execution of a prepared keypress. The chronometric literature conventionally decomposes simple reaction time (~250 ms total for visual stimuli) into premotor and motor components, with the motor (execution) component around 70–100 ms and the remainder perceptual-decisional; since our accumulator absorbs the decisional part but not stimulus-to-cortex transduction, a value between the pure motor component and full simple RT is defensible. I propose **$\lambda = 0.25$ s, with a sensitivity re-fit at 0.15 and 0.35 s** reported alongside — the sensitivity bracket matters more than the point choice, because λ trades one-for-one with the population mean of c and not at all with its spread. (*Reference status, per house rules: the 70–100 ms motor-component and ~250 ms simple-RT figures are textbook-level values I have not verified against a source from this machine; they should be cited properly — Luce's response-time monograph is the standard citation — before any manuscript uses them.*)

5.4 Sequence design: excluded, no simulation. Per Jonas, leaning against; the new data makes it moot. Decision 8 stands unchanged.

6 Order of execution and gates

step	depends on	effort	decides
A.1 recovery harness	—	small	that the estimator works
A.2 hierarchical fit, both bias variants	A.1	small	bias model; population c
A.3 accumulator	A.2	moderate	claim (a): the time gradient
A.4 button validation	A.3	small	paradigm shift; level vs rate
C percentile sensitivity	A.2	small	where the real bottleneck is
B.1 cyclist field + PS check	— (parallel)	small	first transfer evidence
B.2–B.3 truck + LTAP fields	— (parallel)	moderate each	transfer coverage
B.4 the transfer test	A.3 + B.1–B.3	small	claim (b): the gate
D safety audit	A.2, C	moderate	comfort-vs-safety divergence
test-track ingestion	Jonas's data	unknown	second population

My reading of the critical path: A.1 → A.2 → A.3 is the spine; B field constructions can proceed in parallel sessions; nothing above needs data we do not hold except the last row.